

YOUYANG, China

WMO: 576330

Lat: 28.8219N

Lon: 108.7671E

Elev: 828

StdP: 91.77

Time Zone: 8.00 (E08)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-2.6	-1.4	-8.0	2.1	2.5	-6.2	2.5	1.8	4.6	3.3	4.1	4.1	1.2	0	0.320

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	7.5	32.7	23.4	31.5	23.2	30.4	23.0	24.9	29.7	24.4	29.0	24.0	28.3	1.9	140

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
23.7	20.5	26.8	23.2	19.9	26.4	22.8	19.4	26.1	81.4	29.8	79.2	29.2	77.3	28.4	27.1

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
4.2	3.5	3.1	-3.8	34.7	1.9	1.1	-5.2	35.5	-6.3	36.2	-7.3	36.8	-8.7	37.6		
			-4.4	25.7	1.4	0.7	-5.4	26.2	-6.2	26.6	-7.0	27.0	-8.1	27.5		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	15.5	4.2	6.2	10.4	15.7	19.7	22.8	25.7	25.4	21.6	16.4	11.6	6.2
	DBStd	8.00	3.28	3.86	3.93	3.53	2.69	2.47	1.97	2.30	3.01	3.09	3.41	2.87
	HDD10.0	486	181	117	43	3	0	0	0	0	0	1	22	120
	HDD18.3	1798	437	341	247	93	17	1	0	0	7	76	204	376
	CDD10.0	2507	2	9	56	174	300	384	487	478	347	199	68	3
	CDD18.3	777	0	0	1	14	59	135	229	220	103	16	1	0
	CDH23.3	5540	0	1	12	96	343	799	1859	1756	599	72	2	0
CDH26.7	1788	0	0	2	16	76	218	651	657	160	8	0	0	
Wind	WSAvg	1.1	1.1	1.1	1.2	1.1	1.1	1.1	1.1	1.2	1.1	1.0	1.0	1.0
Precipitation	PrecAvg	1332	30	39	64	122	202	237	191	150	98	111	65	22
	PrecMax	1733	57	73	113	206	328	561	396	286	250	204	160	55
	PrecMin	958	8	2	25	29	100	58	39	28	9	48	5	2
	PrecStd	211	14	22	22	43	64	111	93	69	47	47	34	14
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	16.2	19.7	25.0	28.7	30.7	32.1	34.1	34.2	32.3	27.8	23.1	16.9
		MCWB	9.8	12.9	16.3	18.7	19.6	21.9	23.8	23.4	22.6	18.8	15.3	9.9
	2%	DB	12.9	15.8	21.3	26.2	28.8	30.6	32.6	32.8	30.2	25.3	20.8	14.4
		MCWB	8.2	10.5	14.5	18.0	19.4	22.5	23.8	23.5	22.2	17.7	14.4	8.7
	5%	DB	10.6	13.4	18.7	23.9	27.1	29.3	31.5	31.6	28.5	23.2	18.6	12.3
		MCWB	6.8	9.7	13.1	17.2	19.4	22.4	23.7	23.4	21.4	17.3	13.6	7.7
	10%	DB	9.0	11.6	16.4	21.6	25.2	27.7	30.2	30.4	26.8	21.3	16.6	10.5
		MCWB	5.9	8.8	12.1	16.6	19.2	22.0	23.6	23.2	20.7	16.7	13.1	7.3
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	11.5	14.1	18.8	21.1	23.2	24.8	25.7	25.6	24.2	21.6	17.7	11.5
		MCDB	14.6	18.5	22.7	25.9	26.9	28.9	30.5	31.2	29.3	24.8	20.9	14.9
	2%	WB	9.0	11.7	15.5	19.6	22.1	24.1	25.1	24.8	23.4	19.6	15.9	9.8
		MCDB	11.8	14.2	19.5	23.6	25.5	27.9	29.9	29.9	28.0	22.5	18.3	12.6
	5%	WB	7.5	10.5	14.0	18.3	21.0	23.6	24.6	24.3	22.7	18.4	14.6	8.6
		MCDB	9.9	12.5	17.6	21.8	23.9	27.0	29.1	29.1	26.7	21.3	17.1	11.4
	10%	WB	6.2	9.1	12.7	17.2	19.9	23.0	24.1	23.8	22.0	17.4	13.4	7.6
		MCDB	8.6	11.4	15.7	20.5	23.2	26.0	28.4	28.4	25.3	20.0	16.1	10.1
Mean Daily Temperature Range	5% DB	MDBR	5.1	5.5	6.7	7.4	7.4	6.8	7.5	7.9	7.0	6.1	6.1	5.7
		MCDBR	10.1	10.7	12.2	12.6	12.3	10.4	10.0	10.4	10.4	10.7	10.9	11.5
		MCWBR	5.6	6.0	5.7	5.2	4.3	3.4	2.7	2.6	3.3	4.0	5.0	6.0
	5% WB	MCDBR	8.6	8.3	9.9	9.5	7.8	7.2	8.2	8.9	8.1	7.4	7.9	9.1
		MCWBR	5.2	5.1	5.4	4.6	4.0	2.9	2.7	2.7	3.0	3.5	4.3	5.2
Clear-Sky Solar Irradiance	taub	0.679	0.705	0.742	0.710	0.619	0.593	0.547	0.599	0.631	0.691	0.695	0.676	
	taud	1.452	1.437	1.416	1.516	1.738	1.840	1.981	1.820	1.706	1.523	1.480	1.463	
	Ebn at Noon	543	583	603	649	715	730	763	719	675	586	528	517	
	Edh at Noon	264	289	312	291	233	210	182	212	230	261	254	251	
All-Sky Solar Radiation	RadAvg	1.48	1.79	2.61	3.21	3.46	3.61	4.64	4.38	3.35	2.30	1.91	1.48	
	RadStd	0.31	0.42	0.28	0.49	0.66	0.58	0.52	0.47	0.45	0.41	0.39	0.33	

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(46)	Station Only	N/A	N/A	N/A	N/A	-0.33	-0.42	+105	+180	N/A	N/A
(47)	Regional (5 neighbors)	N/A	N/A	N/A	+0.48	N/A	N/A	N/A	N/A	+90	N/A

Nomenclature: See separate page